PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: Finsight - AI Financial Advisor

Version: 1.0

1. Project Overview

- **Problem Statement:** Small business owners and solopreneurs often lack the financial expertise and time to analyze their income and expenses effectively. Traditional accounting software requires manual categorization and doesn't proactively provide actionable insights or risk warnings, leaving owners vulnerable to cash flow issues.
- **Target Domain:** Enabling rapid, AI-driven financial analytics, automated transaction categorization, and instant executive reporting for small businesses.

- **Proposed Solution Summary:** Finsight removes the friction of financial management by automatically categorizing transactions and providing a conversational AI advisor. Users can ask questions about their finances in plain English and instantly receive tailored insights, risk forecasts, and executive reports—shifting their role from manual bookkeeper to strategic decision-maker.

2. Background & Business Objective

- **Background of the Problem:** For years, managing business finances followed a tedious, linear pathway. A business owner enters transactions, manually categorizes each one, and attempts to build spreadsheet models to forecast runway. This process requires extensive labor, specialized accounting knowledge, and blocks many entrepreneurs from truly understanding their financial health.
- **Importance of Solving This Issue:** Accelerating the time to gain financial clarity from days to second. By automating categorization and insight generation, small businesses can pivot quickly, identify cost-cutting opportunities, and avoid cash flow crises without needing a dedicated accountant.

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- **Strategic Fit / Impact:** Finsight aligns with the movement toward AI-native business tools. By integrating the advanced GLM-5.1 model, it transforms raw ledger data into a conversational partner, paving the way for autonomous financial management.

3. Product Purpose

3.1. Main Goal of the System: To provide an AI-enabled, proactive partner in financial management and analysis.

3.2. Intended Users (Target Audience):

- Small Business Owners
- Solopreneurs & Freelancers
- Independent Contractors
- Early-Stage Founders

4. System Functionalities

4.1. Description: The system operates as a high-speed financial intelligence engine. Using a conversational interface and automated backend logic, it captures transaction data, categorizes it, identifies trends, and generates comprehensive executive reports and forecasts.

4.2. Key Functionalities:

- **Automated Categorization:** Translates raw transaction descriptions into structured categories (e.g., Rent, Utilities, Salary) using AI.
- **Conversational Financial Advisor:** Facilitates continuous analysis of financial data by allowing users to chat with an AI advisor (powered by GLM) that has full context of their revenue, expenses, and trends.
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4.3. AI Model & Prompt Design

4.3.1. Model Selection: Zhipu AI (ilmu-glm-5.1). This model was chosen for its deep contextual understanding of conversational memory, advanced reasoning for financial data, and its highly reliable json_mode output, which is critical for parsing categorizations and backend reports.

4.3.2. Prompting Strategy: Multi-step agentic prompting with strict JSON enforcement. The system uses targeted system prompts (zero-shot) for specific tasks, injecting live contextual summaries (total revenue, trends, category breakdowns) into the user prompt to ground the AI's analysis in real data**.**

4.3.3. Context & Input Handling: To handle large transaction datasets without overflowing context windows, Finsight summarizes the data before sending it to LLM. It aggregates total revenues, monthly trends, and category totals. For chat, it only retains the last 6 messages of conversation history to keep the context focused and within token limits.

4.3.4. Fallback & Failure Behavior: If the model returns invalid JSON or off-topic responses, the backend catches the parsing error and logs it. It also implements an automatic retry mechanism (up to 2 retries) for 5xx server errors. For critical mathematical data in reports, Finsight relies on actual database values (source of truth) rather than LLM-generated math, eliminating the risk of math hallucination.

5. User Stories & Use Cases

- **User Stories:** "As a small business owner, I want to ask my AI advisor how I can improve my profit margin, so that I receive tailored cost-cutting recommendations without manually analyzing my ledger."
- **Use Cases (Main Interactions):**

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Transaction Categorization Flow: Users upload transactions, then Finsight parses the data, calls the GLM model for classification, and saves the structured data to the database.

Conversational Insight: User asks a specific financial question. Finsight aggregates recent financial summaries and trend data, passes it to the AI alongside the question, and returns a tailored response and actionable recommendations.

Report Generation: User clicks "Generate Report". Finsight triggers an AI synthesis of recent trends, generating scenario warnings and formatted executive summaries.

6. Features Included (Scope Definition)

- AI-powered transaction categorization engine.
- Conversational chatbot with deep financial context awareness.
- Automated, one-click executive report generation.
- Visual analytics dashboard mapping revenue, expenses, and trends.

7. Features Not Included (Scope Control)

- Direct bank API integrations (e.g., Plaid) [Initial Versions rely on manual entry/CSV].
- Implementation of complex multi-entity corporate tax logic.
- Multi-user simultaneous collaboration environments.

8. Assumptions & Constraints

- **LLM Cost Constraint:** Estimated token cost per average user session is kept low. Design decision: Finsight sends aggregated summaries and trend data to the LLM instead of raw, row-by-row transaction histories for every chat message to keep inference costs manageable on a scale.
- **Technical Constraints:** Requires an active internet connection and valid GLM API credentials. Currently limited to a single-user SQLite database backend.
- **Performance Constraints:** AI generation of complex reports requires significant token processing, leading to brief, expected waiting times (latency) during generation.
- **User Input:** The AI's insights are only as accurate as the transaction data inputted by the user.

9. Risks & Questions Throughout Development

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- **Hallucination on Math:** How do we prevent the AI from hallucinating total revenue or profit? *(Mitigation: The system hardcodes the DB-calculated summary math directly into the final report output, bypassing the AI's math abilities).*
- **Categorization Accuracy:** What happens if the AI miscategorized a vague transaction description?
- **Data Privacy:** How are we ensuring that sensitive user financial data sent to the AI model API is secure and not used for unintended model training?

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